

ARS-Avatar: Animatable and Relightable Surfel Avatars with Learnable Ambient Occlusion

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<https://arsavatar.github.io/>



Figure 1. **ARS-Avatar** creates animatable and relightable human avatars with surfels. Given novel poses and illuminations, it reconstructs high-fidelity *appearance*, *geometry*, *BRDF materials*, and *optimizable occlusion* for realistic animation and relighting.

Abstract

Creating animatable and relightable human avatars from multi-view images remains challenging, as pose-dependent deformation, materials, and light visibility are tightly coupled in images. In this paper, we present **ARS-Avatar**, a novel method using surfel representation for high-quality, animatable, and relightable human avatars from multi-view images captured under unknown illumination. We first extract deformation priors from the template mesh and leverage them as additional guidance beyond driving poses, facilitating faithful estimation of surfel attributes. To support relighting, we employ deferred shading to estimate BRDF materials. We further introduce a differentiable screen-space ambient occlusion formulation that enables gradient-based optimization of body-part specific occlusion radii through finite differences, providing an efficient approximation of light visibility that can be jointly optimized with the avatar. Extensive experiments show that ARS-Avatar achieves competitive or improved radiance reconstruction on multiple metrics and consistently outperforms the evaluated PBR relighting baselines, while enabling realistic animation and relighting under novel poses and illuminations.

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1. Introduction

Creating animatable and relightable human avatars remains a fundamental problem in computer graphics, with applications in embodied AI (EAI), VR/AR, virtual try-on, and telepresence. Traditional methods rely on high-quality 3D scans and calibrated illumination [14, 16]. However, modeling such avatars solely from multi-view images captured under unknown illumination remains ill-posed, as geometry, materials, and illumination are tightly coupled.

Recent advances in neural rendering [29, 41] have substantially improved high-fidelity human avatar reconstruction from multi-view images. Although NeRF-based methods [37, 57, 59, 61] achieve impressive quality, expensive volume rendering limits their practicality, which motivates efficient explicit primitives [26]. Building on 3DGS, recent methods [12, 26, 33, 54, 63, 66] reconstruct animatable and relightable avatars by combining primitives with templates [40, 45] and physically-based rendering [5, 47]. However, many existing methods condition primarily on the driving pose, which may not fully capture rich pose-dependent deformation for clothed humans. Incorporat-

ing explicit deformation priors into the optimization can therefore provide complementary guidance beyond the driving pose. Furthermore, although visibility estimation has been explored for meshes [10, 63] and volume rendering [53, 59], extending it to discrete Gaussian surfels is non-trivial. These limitations jointly constrain the achievable fidelity of pose-dependent animation and relighting.

Therefore, we present **ARS-Avatar**, an animatable and relightable human avatar represented by GaussianSurfels [13] that achieves high-fidelity rendering under novel poses and illuminations. Specifically, we employ SMPL-X [45] to construct the canonical avatar and reconstruct appearance, geometry, and BRDF materials under a driving pose. The driving pose is encoded into a position map [35] and fed into StyleUNets to estimate surfel attributes. We further extract two deformation priors from the template mesh, i.e., the deformation displacement and the deviation from the parametric body model, both of which reflect local deformation complexity. The extracted priors are incorporated into surfel attribute estimation to produce more faithful pose-dependent details. Furthermore, we employ screen-space ambient occlusion (SSAO) as an efficient approximation and incorporate it into the optimization via a finite-difference formulation with respect to the occlusion radius. Each human body part has an initial occlusion radius and a learnable radius offset, and the radii are optimized via finite differences. This formulation enables end-to-end optimization of occlusion while maintaining training efficiency.

In summary, our contributions include:

- We introduce a deformation prior extraction method that computes two vertex-wise priors from the template mesh and incorporates them into pose-dependent surfel attribute estimation for more faithful reconstruction.
- We propose a differentiable screen-space ambient occlusion formulation that optimizes body-part-specific occlusion radii through finite differences, enabling occlusion approximation to be jointly optimized with the avatar representation.
- Extensive experiments on AvatarReX and ActorsHQ datasets demonstrate that **ARS-Avatar** achieves competitive or improved reconstruction quality and enables realistic animation and relighting.

2. Related Work

2.1. Animatable Human Avatar Modeling

Early methods rely on explicit meshes [2, 3, 6, 60] and template body models such as SMPL [40] and SMPL-X [45] to reconstruct and deform avatars. Later methods introduce learnable deformation fields to capture pose-dependent deformations [17, 58]. More recent NeRF-based methods [9, 31, 34, 39, 52, 68, 69] model complex geometry and appearance for continuous, high-fidelity reconstruc-

tion and novel pose animation, but ray sampling limits efficiency. Explicit 3DGS [29] has thus emerged as an effective representation for animatable avatars [19, 20, 25, 30, 32, 35, 43, 46, 67, 71]. Despite efficient rendering, modeling complex clothed humans remains challenging, as conditioning solely on the driving pose does not fully capture rich pose-dependent deformation. To address this, we propose a deformation prior extraction and incorporation method that computes deformation displacement and parametric-body deviation from the template mesh. The deformation priors are further utilized as auxiliary for surfel attribute estimation and improve pose-dependent reconstruction fidelity.

2.2. Relightable Human Avatar Modeling

Relightable avatar modeling is an inverse rendering problem that disentangles geometry, materials, and illumination. Early works use neural networks to relight single images [24, 27, 42, 49, 56, 70], but provide limited support for animation. Chen and Liu [11] pioneered integrating physically-based rendering [50] into NeRF for human geometry and materials. Subsequent methods [22, 59] improve appearance modeling for detailed relighting, while others [7, 37, 53, 61] use explicit light transport for accurate intrinsic decomposition and physically plausible relighting. Recent Gaussian avatars combine explicit primitives with physically based material and illumination models [12, 26, 33, 48, 54, 62, 63], enabling efficient and high-fidelity relighting. We follow these works and explore Gaussian Surfels [13] as the representation primitive, leveraging their surface-aligned advantage for high-fidelity appearance and geometry.

2.3. Visibility Approximation for Avatars

Light visibility is crucial in physically-based rendering, determining light transport between light and surface points. The common approach computes visibility via ray tracing on meshes [10], which typically requires high-quality meshes. Furthermore, NeRF-based methods encode visibility into MLPs [59, 61], and Lin *et al.* [37] divide the body into parts to better model self-occlusion. Recently, visibility for explicit Gaussians has gained attention [15]. One strategy combines Gaussians with a mesh and uses mesh visibility [63]. Liang *et al.* [36] model visibility using ambient occlusion (AO) and encode it into spherical harmonics for efficient querying, later extended to dynamic avatars by Jiang *et al.* [26]. Chen *et al.* [8] adopt screen-space ambient occlusion (SSAO) to improve efficiency of occlusion computation. However, these approaches either precompute visibility or treat it as a separate module, which makes it difficult to jointly optimize visibility with the avatar representation. In contrast, our method formulates SSAO with respect to the occlusion radius through finite differences, enabling end-to-end occlusion optimization while maintaining the ef-

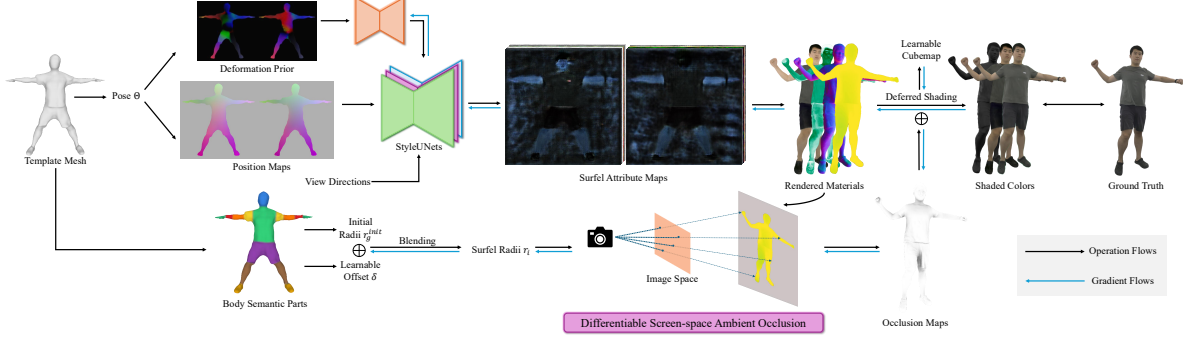


Figure 2. **An illustration of our method.** Given a canonical template mesh, we first deform it into the observation space under a driving pose Θ to generate position maps and extract deformation prior maps based on deformation priors. Both are fed into multiple StyleUNets to estimate attributes of each surfel. We then employ deferred shading to estimate material properties for relighting under novel illumination. Finally, we introduce a differentiable screen-space ambient occlusion (SSAO) module that efficiently estimates occlusion and enables joint optimization with the avatar representation during training.

iciency required for practical avatar reconstruction.

3. Method

3.1. Overview

Given multi-view images of a clothed human, we aim to create an animatable and relightable avatar. As shown in Fig. 2, we represent the avatar with Gaussian Surfels [13]. Following [35], we estimate pose-dependent surfel attributes from position maps using StyleUNets [51]. To better capture the rich deformation features, we extract deformation priors from the template and incorporate them into estimation (Sec. 3.2). We further employ deferred shading to estimate BRDF materials (Sec. 3.3). Light visibility is approximated as screen-space ambient occlusion (SSAO), made differentiable via a finite-difference formulation with respect to the occlusion radius, enabling efficient radius optimization while incorporating visibility into training (Sec. 3.4). Finally, **ARS-Avatar** creates animatable and relightable human avatars under any target pose, viewpoint, and illumination.

3.2. Surfel Estimation with Deformation Priors

We initialize the canonical avatar with surfels $\mathcal{G}$, each parameterized by a position $\mu_c \in \mathbb{R}^3$, rotation $\mathbf{r}_c \in \mathbb{R}^4$, scale $\mathbf{s} \in \mathbb{R}^3$, opacity $\mathbf{o} \in \mathbb{R}^1$, and directional color $\mathbf{c} \in \mathbb{R}^3$. For physically-based shading, we additionally define albedo $\mathbf{a} \in \mathbb{R}^3$ and roughness $\rho \in \mathbb{R}^1$. Given a coarse template mesh and driving pose Θ , we follow Li *et al.* [35] and rasterize template vertices x_c colored by posed coordinates x_o from front and back views to obtain position maps:

$$\mathbf{M}_{pos}^f(\Theta), \mathbf{M}_{pos}^b(\Theta) = \text{Raster}(x_c, x_o; \Theta), \quad (1)$$

where $\text{Raster}(\cdot; \Theta)$ rasterizes the template mesh under pose Θ from front and back views. For deformation priors, we

use two vertex-wise priors: a) the distance before and after deformation $\mathbf{d}$, and b) the distance to the nearest SMPL-X [45] vertex $\mathbf{s}$, both positively correlated with deformation complexity. We render them similarly to obtain deformation prior maps:

$$\mathbf{M}_{def}^f(\Theta), \mathbf{M}_{def}^b(\Theta) = \text{Raster}(\mathbf{x}_c, \{\mathbf{d}, \mathbf{s}\}; \Theta), \quad (2)$$

where $\{\cdot\}$ denotes concatenate operation and we discuss it in the appendix. We encode these maps with a neural network to generate deformation feature maps,

$$f = \mathcal{F}(\mathbf{M}_{def}^f(\Theta), \mathbf{M}_{def}^b(\Theta)), \quad (3)$$

where $\mathcal{F}(\cdot)$ is the neural network. The feature maps are further used as auxiliary to incorporate with position maps into StyleUNets to estimate surfel attributes,

$$\mu_c, \mathbf{r}_c, \mathbf{s}, \mathbf{o}, \mathbf{c}, \mathbf{a}, \rho = \mathcal{S}(\mathbf{M}_{pos}^f, \mathbf{M}_{pos}^b; f, \mathcal{V}), \quad (4)$$

where $\mathcal{V}$ denotes view feature map employed in [35] and $\mathcal{S}(\cdot)$ is the StyleUNets. We further deform canonical surfels to observation space via Linear Blend Skinning (LBS),

$$\mu_o = \mathcal{R}\mu_c + \mathcal{T}, \quad \mathbf{r}_o = \mathcal{R}\mathbf{r}_c, \quad (5)$$

where $\mathcal{R}$ and $\mathcal{T}$ are rotation and translation from skinning weights. The covariance matrix is:

$$\Sigma_o = \mathbf{r}_o \mathbf{s} \mathbf{s}^T \mathbf{r}_o^T, \quad (6)$$

finally, we render the avatar to obtain the rendered color $\hat{C}$, depth $\hat{D}$, normal $\hat{N}$, albedo $\hat{A}$, roughness $\hat{H}$, and position $\hat{P}$,

$$\hat{C}, \hat{D}, \hat{N}, \hat{A}, \hat{H}, \hat{P} = \text{Splat}(\mathbf{c}, \mathbf{a}, \rho; \mu_o, \Sigma_o, \mathbf{o}), \quad (7)$$

where $\text{Splat}(\cdot; \cdot)$ denotes EWA splatting [72] adapted from [13] and extended to render additional feature maps. We further extend the formulation in Sec. 3.4.

3.3. Material Estimation with Deferred Shading

We follow the rendering equation [47] for physically-based shading,

$$L_o(\omega_o) = \int_{\Omega} L_i(\omega_i) f_r(\omega_i, \omega_o) (\omega_i \cdot \mathbf{n}) d\omega_i, \quad (8)$$

where $L_o(\omega_o)$ and $L_i(\omega_i)$ are outgoing radiance in direction $\omega_o = P_{\text{cam}} - \hat{P}$ and incident light from ω_i . The incident light is represented by a learnable HDR cube map with the shape $6 \times 512 \times 512$. We use the Disney shading model [5] with albedo $\mathbf{a}$, and roughness ρ as BRDF materials. We decompose the BRDF term f_r into diffuse f_{diff} and specular f_{spec} . The shaded color is then computed as,

$$\hat{C}_{\text{shade}} = C_{\text{diff}} + C_{\text{spec}}, \quad (9)$$

we discuss the definitions of these two terms in the appendix. Combining with ambient occlusion $\mathcal{O}$ discussed in Sec. 3.4, the physically-based shading color is:

$$\hat{C} = \mathcal{O} \odot \hat{C}_{\text{shade}}, \quad (10)$$

where $\odot$ denotes element-wise multiplication. The learnable cube map L_i is optimized jointly with surfels to estimate incident illumination from observations.

3.4. Differentiable Screen-space Ambient Occlusion

We approximate light visibility with screen-space ambient occlusion (SSAO) [4], which calculates local visibility from the rendered depth $\hat{D}$ and normal $\hat{N}$. For each pixel p ,

$$\text{SSAO}(p) = \frac{1}{N} \cdot \sum_{i=1}^N V(p, \omega_i), \quad (11)$$

where $V(p, \omega_i) \in [0, 1]$ is visibility along sampled direction ω_i . However, standard SSAO is not directly amenable to end-to-end optimization, and optimizing its input depth and normal maps would implicitly modify the underlying geometry and surface orientation.

We therefore introduce a dedicated optimizable path that only adjusts occlusion radius while keeping geometry inputs fixed. We divide the human body into nine semantic parts, each with an initial radius and a learnable scalar offset. For part g :

$$r_g = r_g^{\text{init}} (1 + \delta_g), \quad g = 1, \dots, 9, \quad (12)$$

the radius associated with each surfel is obtained by blending the radii of its neighboring joints according to the LBS weights:

$$r_i = \sum_{j=1}^J w_{ij} r_{g(j)}, \quad (13)$$

where w_{ij} denotes the LBS weight of surfel i for joint j , and $g(j)$ maps joint j to its corresponding body part.

We generate perturbed radii r_g^+ and r_g^- by adding a perturbation offset $\varepsilon \Delta_g$ to the corresponding body parts, where $\Delta_g \in \{-1, +1\}$ and ε is a small perturbation magnitude. We splat above 3 radii to obtain radius map $\hat{R}, \hat{R}_+, \hat{R}_-$ via Eq. 7. The final and perturbed occlusion map is computed as:

$$\hat{O}, \hat{O}_+, \hat{O}_- = \mathcal{A}(\hat{N}, \hat{D}; \hat{R}, \hat{R}_+, \hat{R}_-), \quad (14)$$

where $\mathcal{A}$ denotes SSAO generation function. Therefore, we estimate the gradient via the finite difference formulation,

$$\frac{\partial \hat{O}}{\partial \hat{R}} \approx \frac{\hat{O}_+ - \hat{O}_-}{\hat{R}_+ - \hat{R}_-}, \quad (15)$$

since PyTorch [44] and the EWA splatting [72] are differentiable, the full gradient for radius compensation is:

$$\frac{\partial \mathcal{L}}{\partial \delta} = \frac{\partial \mathcal{L}}{\partial \hat{O}} \cdot \frac{\partial \hat{O}}{\partial \hat{R}} \cdot \frac{\partial \hat{R}}{\partial r_i} \cdot \frac{\partial r_i}{\partial \delta_g}, \quad (16)$$

finally, the final shading result is obtained by:

$$\hat{C}_{\text{pbr}} = C_{\text{shade}} \cdot \hat{O}, \quad (17)$$

we provide a more detailed explanation in the appendix.

3.5. Loss Function and Optimization

We optimize ARS-Avatar using multi-view RGB images C . The training losses combines L1, perceptual, geometry, normal deformation, light, BRDF, and regularization terms:

$$\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_{\text{lpips}} + \mathcal{L}_{\text{geo}} + \mathcal{L}_{\text{nd}} + \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{TV}} + \mathcal{L}_{\text{light}}, \quad (18)$$

where the geometry loss enforces consistency between rendered normal and depth map. We further propose $\mathcal{L}_{\text{nd}}$ to regularize normal deformation across neighboring surfels,

$$\mathcal{L}_{\text{nd}} = \frac{1}{|\mathcal{K}|} \sum_{(i,j) \in \mathcal{K}} w_{ij} (1 - \mathbf{n}_{oi} \cdot \mathbf{n}_{oj}), \quad (19)$$

where $\mathbf{n}_{oi}$ denotes the observation normal of i -th surfel, and $\mathcal{K}$ contains selected neighboring pairs satisfying the canonical distance constraint. w_{ij} is a spatial weight based on observation space distance between surfels i and j , which downweights distant pairs. Its exact form and definition of the local scale σ_i are provided in the appendix.

4. Experiments

Dataset. The experiments are conducted on Avatar-ReX [69] and ActorsHQ [23] datasets. We follow the setting in [35] and select 4 characters from AvatarReX and 2 actors from ActorsHQ dataset for training.



Figure 3. **Illustration of appearance and geometry reconstruction, BRDF materials estimation, and relighting synthesis under training and novel poses.** The second column shows the results of directly fitting Surfel colors, while the third column shows the results obtained using physically-based rendering.

Metric. We employ Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM) [55], Learned Perceptual Image Patch Similarity (LPIPS) [65], and Frechet Inception Distance (FID) [18] as metrics.

Baseline. We compare radiance reconstruction against 3DGS-Avatar [46], WebAvatar [64], and AnimatableGaussian [35], and further compare physically-based shading and material estimation against MeshAvatar [10] and Relighting4D [11].

Results. We report animation and relighting results of clothed humans created by ARS-Avatar in Fig. 1 and Fig. 3.

4.1. Comparison

Animatable Avatar Reconstruction. We first evaluate the animation capability of ARS-Avatar against

the evaluated avatar reconstruction methods, including 3DGS-Avatar [46], AnimatableGaussians [33], and WebAvatar [64]. Qualitative and quantitative results are reported in Fig. 4 and Tab. 1. ARS-Avatar achieves competitive metric results and high-fidelity visual results under novel poses, especially in regions with complex nonrigid deformation, such as loose garments and articulated body parts. In contrast, existing methods generally rely on rigid deformation and driving pose information, which may fail to capture rich pose-dependent details of clothed humans.

Relightable Avatar Reconstruction. We further compare the relighting capability of RAS-Avatar with recent physically-based avatar reconstruction methods, including MeshAvatar [10], and Relighting4D [11]. Qualitative and quantitative comparisons are reported in Fig. 5 and Tab. 2,

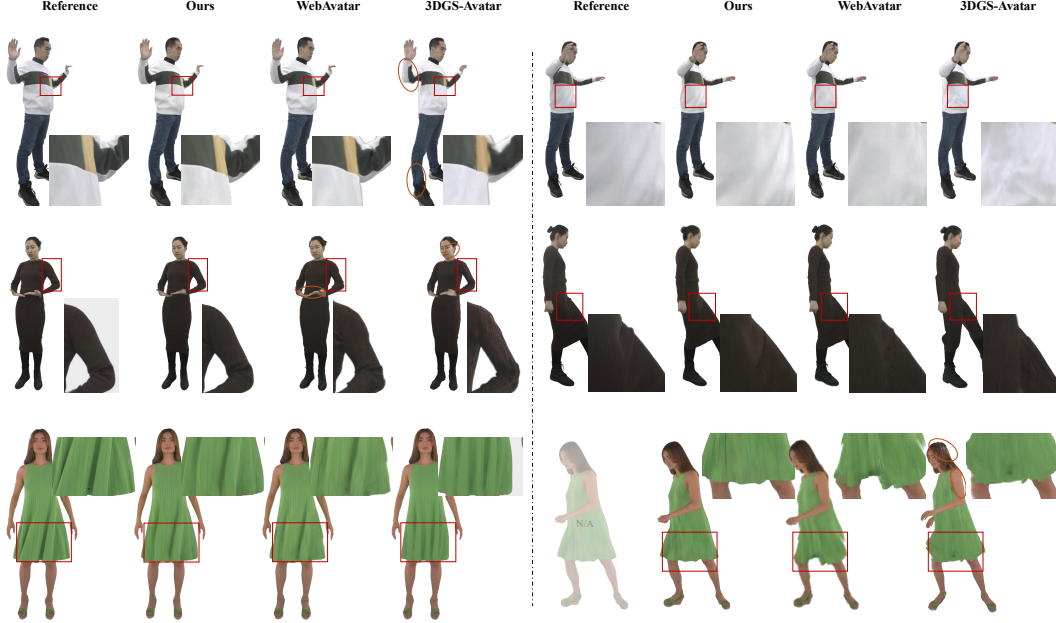


Figure 4. **Qualitative Radiance Comparison** with evaluated human avatar reconstruction methods including 3DGS-Avatar [46] and WebAvatar [64] under testing and novel poses.

Table 1. **Quantitative Comparison against state-of-the-art radiance reconstruction baseline on AvatarReX [69] and Actor-sHQ [23] dataset.** All methods are evaluated under the same settings. *Results of AG [35] are from their paper.

	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$
REX	3DA [46]	29.2612	0.9729	0.0389	29.6184
	AG* [35]	30.6143	0.9803	0.0290	13.2417
	WEB [64]	31.6588	0.9809	0.0297	23.1407
	Ours	<u>30.7514</u>	0.9862	0.0318	13.1050
HQ	3DA [46]	30.1744	0.9567	0.0317	16.7763
	AG* [35]	30.3607	0.9682	0.0339	33.4665
	WEB [64]	<u>31.3856</u>	<u>0.9686</u>	0.0324	31.5130
	Ours	31.4785	0.9694	0.0299	<u>18.9706</u>

Table 2. **Quantitative Comparison against the evaluated PBR baselines on AvatarReX [69] and ActorsHQ [23] dataset.** ARS-Avatar outperforms the baselines on the evaluated metrics.

	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$
REX	R4D [11]	23.6745	0.9252	0.0974	96.4823
	MA [10]	28.5608	0.9707	0.0377	18.2602
	Ours	30.2703	0.9752	0.0375	12.7732
HQ	R4D [11]	24.1780	0.9457	0.0627	139.2214
	MA [10]	28.7860	0.9510	0.0305	19.7140
	Ours	31.6116	0.9596	0.0261	17.1087

respectively. ARS-Avatar produces more consistent shad-

Table 3. **Quantitative Ablation on Deformation Priors (DP).**

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$
Ours w/o MPG	30.9457	0.9515	0.0446	34.5386
Ours	31.7459	0.9560	0.0382	33.4016

ing and occlusion effects than the evaluated PBR baselines under novel lighting. Although explicit mesh or neural field methods achieve physically based appearance modeling, their visibility estimation relies on explicit geometry processing or expensive light transport. In contrast, Our introduced learnable screen-space ambient occlusion formulation enables efficient visibility approximation and joint optimization with the avatar. This yields more consistent occlusion and contact visibility without costly ray tracing or offline computation.

4.2. Ablation Study

We conduct extensive ablation studies to evaluate the effectiveness of our proposed components.

Effect of Deformation Priors. We compare the model without deformation priors with our full model, where pose-dependent surfel attributes are directly predicted from the position map. As shown in Fig. 7, introducing deformation priors improves all evaluated metrics by providing additional spatial guidance for learning pose-dependent offsets. This enables the model to better preserve fine-grained pose-dependent details, especially in regions with large de-



Figure 5. **Qualitative Physical Comparison** with evaluated human avatar relighting methods including meshavatar [10] and relighting4D [11] under testing and novel poses.

formations, such as the limbs.

Effect of Differentiable Screen-space Ambient Occlusion. We validate the results and time costs with five variants in Fig. 6. The overhead introduced by the learnable radius is negligible, as the addition operations involving

the radius account for only a small fraction of the total time. Compared with RAGA-style occlusion, which is post-computed and cannot be jointly optimized with the avatar, our method produces more accurate occlusion maps, especially in self-occluded regions. As shown in Fig. 6, com-

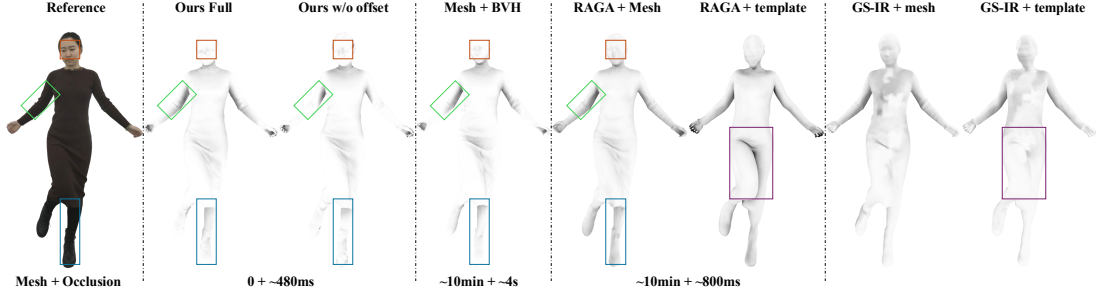


Figure 6. **Qualitative Ablation on Differentiable Screen-space Ambient Occlusion.** We compare five variants: (1) our full method; (2) our method without the optimizable radii offsets; (3) mesh-based [13] occlusion accelerated by BVH [28]; (4) RAGA style occlusion [63]; (5) GS-IR style occlusion [36]. Our method produces high-quality occlusion results in real time without additional operation. Time for mesh extraction and occlusion computation is reported below. Variant (5) is omitted from the time comparison due to unstable results.

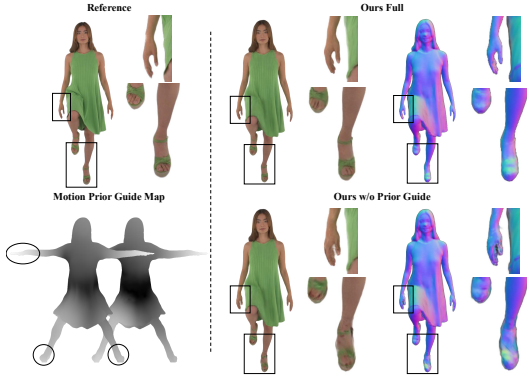


Figure 7. **Qualitative Ablation on Deformation Priors.** Our deformation prior focuses on regions with significant deformation, such as the hands and feet. Incorporating it into surfel estimation yields more refined reconstructions in these regions.

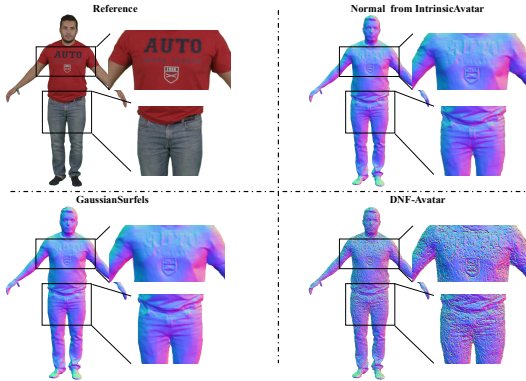


Figure 8. **Qualitative Ablation on Surfel Representation.** Compared with [21], gaussiansurfel [13] representation reconstructs more accurate depth. We convert the depth into normal for clearer comparison. Training process is based on [26] and normal supervision is derived from [53].

extraction and ray tracing, our method achieves comparable occlusion accuracy with significantly less time. Therefore, considering the improved occlusion accuracy and the ability to jointly optimize visibility with the avatar, the overhead of our differentiable SSAO is acceptable.

Effect of Gaussian Surfel Representation. Unlike conventional 3D Gaussian primitives, Gaussian Surfels explicitly encode surface orientation via surface-aligned anisotropic primitives, providing more reliable geometry for physically-based rendering. As shown in Fig. 8, they produce more coherent depth and normal maps with sharper geometric boundaries than conventional Gaussians, which benefit material estimation and visibility modeling and leads to more realistic relighting.

5. Conclusion

We present **ARS-Avatar**, which creates animatable and relightable human avatars from multi-view images. Surfel primitives achieve high fidelity and consistency in appearance and geometry with faster rendering than implicit methods. We further propose a differentiable ambient occlusion method for shadow estimation in the 3DGS pipeline, which can be integrated into optimization without additional network components while maintaining efficiency. Our method offers an effective solution for animatable and relightable 3D human avatar modeling.

However, It is undeniable that the network requires a long time to complete optimization, and our deformation prior extraction relies on predefined priors, which not capture diverse information across clothed human body. Future work could incorporate richer physical knowledge or priors from large-language model [1, 38]. Moreover, while our current SSAO only optimizes body-part-specific occlusion radii, it could be integrated into StyleUNets so that all SSAO-related parameters, rather than only the radius, are estimated and jointly optimized with the avatar. We leave this exploration for future work.

pared with mesh-based AO, which requires additional mesh

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Supplemental Document

6. Implementation Details

All models are implemented in PyTorch and trained on a single NVIDIA RTX 3090 GPU with 24 GB of memory. The training iterations for radiance reconstruction and physically-based shading estimation are set to 500k and 300k, with training times of approximately 4 and 2 days, respectively. The rasteriser, the cubemap filtering and the differentiable ambient occlusion formulation are custom CUDA extensions. The split-sum pre-filtering uses the same cosine and GGX kernels as the public `nvdiffrrec` release. The PCA operation for out-of-distribution pose [35] is also performed on position map and deformation priors map for more faithful reconstruction.

7. Detailed Deformation Feature Maps

As mentioned in Sec. 3.2, we define the deformation feature maps by utilizing two distance: a) the distance before and after deformation $\mathbf{d}$, and b) the distance to the nearest SMPL-X [45] vertex $\mathbf{s}$. For the two distances, we encode their magnitude and orientation into the feature map $\mathbf{M}_{def}^f$ and $\mathbf{M}_{def}^b$ respectively. However, for distance $\mathbf{s}$, it measures the error of the weights in the LBS deformation process. Therefore, we believe that its magnitude is more valuable. Therefore, the final encoded information includes the magnitude and orientation of $\mathbf{d}$ and the magnitude of $\mathbf{s}$,

$$\{\mathbf{d}, \mathbf{s}\} = \text{Concate} \left(\|\mathbf{d}\|, \frac{\mathbf{d}}{\|\mathbf{d}\|}, \|\mathbf{s}\| \right), \quad (20)$$

where `Concate` denotes concatenate operation.

8. Detailed Splatting Formulation

We employ `Splat(·; ·)` introduced in GaussianSrfel [13], which render color $\hat{C}$, normal $\hat{N}$, depth $\hat{D}$ maps from a set of surfel. We extend the formulation in manuscript to render albedo $\hat{A}$, roughness $\hat{H}$, position $\hat{P}$, radius $\hat{R}$, perturbed radius $\hat{R}_+$, $\hat{R}_-$, sample importance I , and SSAO bone-aware direction Q maps. Therefore, the final splatting formulation is defined as:

$$\hat{C}, \hat{D}, \hat{N}, \hat{A}, \hat{H}, \hat{P}, \hat{R}, \hat{R}_+, \hat{R}_-, I, Q = \text{Splat}(\mathbf{c}, \mathbf{a}, \rho; \mu_o, \Sigma_o, \mathbf{o}), \quad (21)$$

we will discuss the sample importance map I and SSAO direction map Q in Sec. 10. The rendered albedo and roughness map is splatted without the influence of background, which avoids training failure caused by background noise during deferred shading.

9. Detailed Physically-Based Shading

Specifically, we decompose the BRDF term into diffuse and specular components:

$$f_r(\omega_i, \omega_o) = f_{\text{diff}}(\omega_i) + f_{\text{spec}}(\omega_i, \omega_o), \quad (22)$$

The diffuse color is computed by:

$$C_{\text{diff}} = \frac{\mathbf{a}}{\pi} \cdot \int_{\Omega} L_i(\omega_i)(\omega_i \cdot \mathbf{n}) d\omega_i, \quad (23)$$

For specular component, we utilize SplitSum approximation to separate the light and BRDF term in the integrals. Therefore, the specular color is computed by:

$$C_{\text{spec}} \approx \int_{\Omega} L_i(\omega_i) D(\omega_i, \omega_o)(\omega_i, n) d\omega_i \cdot \int_{\Omega} f_{\text{spec}}(\omega_i, \omega_o) \cdot (\omega_i \cdot n) d\omega_i \quad (24)$$

where $D(\omega_i, \omega_o)$ denotes normal distribution function. We discuss the terms D and f_{spec} in the appendix. The diffuse and specular components are finally combined to obtain the shaded color before accounting for visibility:

$$\hat{C}_{\text{shade}} = C_{\text{diff}} + C_{\text{spec}}, \quad (25)$$

As mentioned in Sec. 3.3, we employ the rendering equation [47] on rendered maps, and the BRDF term $f_r(\omega_i, \omega_o)$ is decomposed into diffuse and specular components. The diffuse component is defined as:

$$f_{\text{diff}}(\omega_i) = (1 - m) \cdot \frac{\mathbf{a}}{\pi}, \quad (26)$$

We set metallic to 0 for all surfels, as we focus on diffuse-dominant human skin and clothing. Hair and accessories with strong specularity may require a metallic channel, which we leave for future work. We simplify the Eq. 26 by set metallic equals 0 to get the final diffuse BRDF term $f_{\text{diff}} = \frac{\mathbf{a}}{\pi}$.

For specular component f_{spec} , we define it using normal distribution function D , Fresnel term F , and geometry term G , which are calculated in [50],

$$f_{\text{spec}}(\omega_i, \omega_o) = \frac{D \cdot F \cdot G}{4 \cdot (\omega_i \cdot n) \cdot (\omega_o \cdot n)}, \quad (27)$$

where n denotes the normal of current pixel.

10. Detailed Differentiable Ambient Occlusion

This section provides the detailed formulation of the differentiable SSAO module introduced in Sec. 3.4. The central objective is to enable gradient-based optimization of the occlusion radius while preserving the efficient CUDA implementation of the original SSAO operator.

10.1. Body-Part Specific Radius Parameterization

We divide the human body into nine semantic parts and generate a body set:

$$\mathcal{B} = \{\text{torso, hip, knee, foot, head, shoulder, elbow, wrist, hand}\}, \quad (28)$$

each body part $b \in \mathcal{B}$ is assigned an initial radius r_g^{init} and a learnable offset δ_g . We have discussed how to obtain the radius of each surfel in Eq. (12 ~ 15). Finally, we assign the radius attribute r_i to each surfel and make it participate in splatting to obtain the rendered radius map $\hat{R}$.

10.2. Custom Backward Propagation

Given an upstream gradient $\frac{\partial \mathcal{L}}{\partial \hat{O}}$, the custom backward operation computes as,

$$\frac{\partial \mathcal{L}}{\partial \hat{R}} = \frac{\partial \mathcal{L}}{\partial \hat{O}} \cdot \frac{\partial \hat{O}}{\partial \hat{R}}, \quad (29)$$

the resulting gradient is propagated through the differentiable baseline radius branch:

$$\frac{\partial \mathcal{L}}{\partial \hat{R}} \rightarrow \frac{\partial \mathcal{L}}{\partial \{r_i\}} \rightarrow \frac{\partial \mathcal{L}}{\partial \{r_g\}} \rightarrow \frac{\partial \mathcal{L}}{\partial \{\delta_g\}_{g=1}^9}, \quad (30)$$

two perturbed positive and negative radius branches are detached from the computational graph and therefore do not receive gradients. Likewise, the normal, depth, bone-aware direction, and importance maps are treated as fixed inputs to the differentiable SSAO module:

$$\frac{\partial \hat{O}}{\partial \hat{N}} = \frac{\partial \hat{O}}{\partial \hat{D}} = \frac{\partial \hat{O}}{\partial \hat{Q}} = \frac{\partial \hat{O}}{\partial \hat{I}} = \mathbf{0}, \quad (31)$$

consequently, the proposed module provides a gradient path only for the body-part specific radius parameters.

10.3. Bone-Aware Direction Map

To guide the SSAO sampling direction, we introduce a bone-aware direction map Q that incorporates pose-dependent skeletal information into the sampling process. For each surfel primitive i , we first compute its bone-aware direction by blending the joint frame axes with the corresponding LBS weights:

$$\mathbf{q}_i^{\text{bone}} = \text{normalize} \left(\sum_{j=1}^J w_{ij} \mathbf{V}_j \mathbf{e}_y \right), \quad (32)$$

where w_{ij} denotes the LBS weight between surfel i and joint j , $\mathbf{e}_y = (0, 1, 0)$ is the canonical up axis, and $\mathbf{V}_j = \mathbf{R}_j \mathbf{R}_j^{\text{rest}^{-1}}$ is the pose-dependent rotation of joint j , expressed as the joint's canonical-to-live rotation $\mathbf{R}_j$ composed with the inverse of its rotation $\mathbf{R}_j^{\text{rest}}$ in the canonical pose.

The direction $\mathbf{q}_i^{\text{bone}}$ is produced in world space and is rotated into camera space before being splatted, so that it is expressed in the same frame as the camera-space surfel normal $\mathbf{n}_i$. Since SSAO sampling is performed around the surface tangent plane, we project the bone direction onto the tangent space of each surfel:

$$\mathbf{t}'_i = \mathbf{q}_i^{\text{bone}} - (\mathbf{q}_i^{\text{bone}} \cdot \mathbf{n}_i) \mathbf{n}_i, \quad (33)$$

$$\mathbf{t}_i = \text{normalize}(\mathbf{t}'_i), \quad \mathbf{b}_i = \mathbf{n}_i \times \mathbf{t}_i, \quad (34)$$

the tangent basis is then defined as:

$$\mathbf{Q}_i = [\mathbf{t}_i, \mathbf{b}_i, \mathbf{n}_i], \quad (35)$$

Similar to other surfel attributes, the direction field $\{\mathbf{q}_i^{\text{bone}}\}$ is splatted into image space using $\text{Splat}(\cdot; \cdot)$ to obtain the bone-aware direction map Q .

10.4. Sample Importance Map

Our screen-space ambient occlusion solver marches a fixed set of hemisphere directions on a uniform angular grid, so that every pixel receives the same number of occlusion rays. This uniform allocation is a poor fit for an articulated human body, whose parts span more than an order of magnitude in size: because a ray-marching step maps to a screen-space displacement proportional to the surface depth, a density tuned to resolve occlusion on the trunk wastes samples on its wide variation, while the same density is far too sparse to resolve the narrow occlusion features of fingers and wrists. We therefore introduce a sample importance map I that allocates SSAO sampling density according to the local body-part scale, and is cheap to evaluate because it reuses the LBS weights and joints already available from skinning.

For each surfel i we first compute its skeletal reference position by LBS-weighted interpolation of joint locations:

$$\mathbf{x}_i^{\text{bone}} = \sum_{j=1}^J w_{ij} \mathbf{p}_j, \quad (36)$$

where $\mathbf{p}_j$ denotes the position of the j -th joint in the posed frame, w_{ij} is the LBS weight of surfel i at joint j , and J is the number of joints.

A uniform distance decay would treat all joints alike, but the relevant length scale differs by an order of magnitude between, e.g., a finger joint and the pelvis. We therefore associate each joint j with a nominal body-part radius r_j and define the importance of a surfel as the ratio between its blended body-part scale and the per-joint scale of the reference position:

$$I_i = \frac{\sum_{j=1}^J w_{ij} e^{-\|\mathbf{x}_i - \mathbf{p}_j\|_2 / \sigma} r_j}{\sum_{j=1}^J w_{ij} r_j} \in [0, 1], \quad (37)$$

where $\mathbf{x}_i$ is the position of surfel i , and $\sigma = 0.15\text{ m}$ is a distance fall-off radius. The numerator is an exponentially distance-weighted average of the per-joint radii over the joints that actually influence the surfel, so a surfel lying on a thin structure such as a finger inherits the small radius of its governing joint and obtains $I_i \rightarrow 1$, whereas a surfel on a thick structure such as the torso is dominated by the large radii of the trunk joints and obtains a small I_i . The exponential stencil makes the transition between neighbouring body parts smooth, so no part segmentation is required; because it appears in both numerator and denominator, the ratio is also independent of the absolute scale of the joint radii.

Similar to other surfel attributes, the importance values are packed into the SSAO feature channel of the surfel and splatted using $\text{Splat}(\cdot; \cdot)$. The obtained importance map I is used to adjust the angular sampling density of SSAO:

$$\delta_{\text{adp}} = \frac{\delta}{1 + \alpha I}, \quad (38)$$

where $\delta = 0.0625$ is the default angular sample interval, and $\alpha = 3$ is a weight that controls the influence of the importance map on the sampling density. Allocating a full unit of angular budget to the most articulated regions yields a denser step $\delta_{\text{adp}} \in [\delta/4, \delta]$, i.e. up to four times more sampled directions on thin appendages, while leaving the sampling on the torso unchanged. The kernel clamps the incoming importance to $[0, 1]$ before applying (38) to keep the number of sampled directions bounded in all cases.

Table 4. **Loss terms and weights.** The depth normal consistency weight ramps up from 0.025 to 0.125 during training.

Loss	Symbol	Weight	Stage
Photometric L1	$\mathcal{L}_1$	1.0	Rad., PBR
Perceptual (LPIPS)	$\mathcal{L}_{\text{lpiips}}$	0.1	Rad., PBR
Offset regularisation	$\mathcal{L}_{\text{reg}}$	0.005	Rad., PBR
Depth normal consistency	$\mathcal{L}_{\text{cons}}$	0.025	Rad., PBR
Curvature smoothness	$\mathcal{L}_{\text{curv}}$	0.005	Rad., PBR
Normal deformation	$\mathcal{L}_{\text{nd}}$	0.004	Rad.
BRDF TV	$\mathcal{L}_{\text{TV}}$	0.005	PBR
Environment prior	$\mathcal{L}_{\text{light}}$	0.05	PBR

10.5. Detailed Training Losses

The training process includes two stage, In the first stage, we use the directional color and reconstruct the geometry of the human avatar. The training loss consist of radiance loss, geometry loss, normal deformation loss, and regularization. The radiance loss includes L1 and lpips loss,

$$\mathcal{L}_{\text{rad}} = \mathcal{L}_1(C, \hat{C}) + \mathcal{L}_{\text{lpiips}}(C, \hat{C}), \quad (39)$$

and geometry loss is defined as:

$$\mathcal{L}_{\text{geo}} = \mathcal{L}_{\text{cons}}(\hat{D}, \hat{N}) + \mathcal{L}_{\text{curv}}, \quad (40)$$

We have discussed the definition of normal deformation loss $\mathcal{L}_{\text{nd}}$ and the spatial weight in manuscript is defined as,

$$w_{ij} = \exp\left(-\frac{\|\boldsymbol{\mu}_{oi} - \boldsymbol{\mu}_{oj}\|_2^2}{\sigma_i^2}\right), \quad (41)$$

where σ_i is the local distance scale from the farthest neighbor in the K -nearest neighborhood of the i -th surfel. We further regularize the offset $\boldsymbol{\mu}$ between the optimized and initial surfel position,

$$\mathcal{L}_{\text{reg}} = \|\Delta\boldsymbol{\mu}\|_2^2, \quad (42)$$

in the second stage, we estimate BRDF materials and light and add extended training loss, including shading loss, TV smooth loss, and regularization of light.

$$\mathcal{L}_{\text{shaded}} = \mathcal{L}_1(C, \hat{C}_{\text{pbr}}) + \mathcal{L}_{\text{lpiips}}(C, \hat{C}_{\text{pbr}}), \quad (43)$$

We employ the TV smooth loss $\mathcal{L}_{\text{TV}}$ on rendered albedo and rendered roughness map and environment light, and enforce the light tend to white light,

$$\mathcal{L}_{\text{light}} = \frac{1}{3} \sum_{i=1}^3 \left\| l_i - \frac{1}{3} \sum_{j=1}^3 l_j \right\|. \quad (44)$$

we use $\mathcal{L}_{\text{rad}}$ in the first stage and switch to $\mathcal{L}_{\text{shaded}}$ in the second stage, while also adding $\mathcal{L}_{\text{TV}}$ and $\mathcal{L}_{\text{light}}$.

11. Ethical Considerations.

Our method can synthesize animatable and relightable human avatars, which could be misused to generate fake videos or non-consensual avatar content. We therefore emphasize that any deployment of this technology should obtain explicit consent from the depicted individuals and should include safeguards such as watermarking and deep-fake detection.